A BERT-Based Approach For Predicting Stack Overflow Question Quality

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Abstract—With the growing popularity of Community Ques- tion Answering (CQA) sites like Stack Overflow, the quality and novelty of posted questions have decreased considerably, resulting in redundancy and extra workload for human moderators. This study further works and suggests a deep learning-based method to automatically rate the quality of Stack Overflow posts. With the publicly available "60k Stack Overflow Questions with Quality Rating" dataset, used under a controlled preprocessing pipeline of data cleaning, text normalization, lemmatization, and stripping of HTML tags and extraneous content. Then we used the preprocessed data to train and fine-tune six models: LSTM, GRU, and four transformer-based models: BERT, DistilBERT, ALBERT, and RoBERTa. The models were evaluated based on traditional metrics. Results show that BERT performed the best among all the models, providing 93% accuracy, surpass- ing prior work in this field. The results of the proposed study prove the effectiveness of transformer architectures in enhancing automated moderation and content quality in online technical forums.

Index Terms—Stack Overflow, Deep Learning, Natural Lan- guage Processing (NLP), BERT, BiLSTM, Transformer Models, Attention Mechanism

I. Introduction

Community based question-answering sites have seen significant growth in their usage due to the increasing number of users [1]. Sites like Stackoverflow, Quora, Reddit have become widespread and commonly used for solving day-to-day problems. The increased traffic and queries also leads to redundancy and duplicacy as the quality of questions asked decreases. Stackoverflow allows users to freely post questions and get help and additional sources related to their topic. By March 2024, Stack Overflow had surpassed 23 million registered users, with more than 24 million questions and 35 million answers [2]. Despite prompts encouraging users to check for existing discussions, the platform still contains a large amount of repeated questions that have already been asked or answered. To cut down the number of duplicates, users with high reputations manually flag the duplicates as such, resulting in the wastage of time effort. On the other hand, newcomers to the field of computer sciences with limited knowledge of their domains, fail to explain their problems in a meaningful way, leading to a decline in quality of questions on the platform. Commonly encountered issues include duplicacy, vagueness and lack of examples and context. These problems were not a concern in the early days of the site but have become more prevalent as their scale has grown over the years. With more than a thousand questions being put on Q&A sites everyday, Must is a need for automated software to replace manual operations that are currently being used [3]. Utilising the attributes of a stackoverflow post, we can precisely predict the question's type and help in automating of the moderation.

This research leverages three key techniques—Natural Lan- guage Processing (NLP), Data Analytics, and Deep Learn- ing—to effectively predict and analyze question quality on platforms like Stack Overflow, along with the use of deep learning models and their variants, which are discussed below.

- BERT (Bidirectional Encoder Representations from Transformers) [4] is a cutting-edge language representa- tion model developed to enhance deep bidirectional text comprehension. Unlike earlier models, BERT represents context from both sides in all layers of unlabeled text, which improves comprehension. Following pre-training, it can be fine-tuned with minimal modification to achieve high performance on a broad spectrum of natural lan- guage processing tasks. The model's success is attributed to its training procedures and architecture, including next sentence prediction and masked language modeling. Its superior performance on the GLUE, SQuAD, and SWAG datasets is evident from comparisons, and other studies highlight bidirectionality, pre-training, and scaling of models.
- DistilBERT is a distilled and lightweight version of the BERT model [5], designed to reduce the size of the base BERT architecture by 40% while retaining 97% of its language understanding capabilities. It achieves a 60% increase in output generation speed, making it apt for environments having limited computational resources while still maintaining high performance on downstream NLP tasks.
- ALBERT (A LITE BERT) is another variant of the BERT model [6],is an optimized version of BERT that re- duces the number of model parameters through factorized embedding parameterization and cross-layer parameter sharing, significantly improving memory efficiency and training speed without compromising performance on language understanding benchmarks.
- RoBERTa (Robustly Optimized BERT Pretraining Ap- proach) [7] is a variant of BERT that improves perfor- mance by training on more data with dynamic

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masking, removing the next sentence prediction objective, and optimizing hyperparameters, resulting in more robust and accurate language representations.

- LSTM (Long Short-Term Memory) neural networks[8] were introduced to deal with the vanishing and exploding gradients problem of traditional RNNs, where gradients become too small or too large, eventually making it im- possible for the model to capture long term dependencies. LSTM solves this problem by introducing a cell state that carries information through time steps with minimal modification. LSTM networks also have 3 gates (forget, input, output) that control the flow of information by deciding what information the model should retain and what it should delete.
- GRU (Gated Recurrent Unit) [9]is a variant of a recurrent neural network that, like LSTM, addresses the challenges of vanishing gradient faced by more tradional versions of RNNs and captures the long range dependencies in sequential data. The GRU is computationally efficient while still maintaining the ability to learn long range relations in sequential data. In this study, we used a GRU model to process the Stack Overflow posts, capturing the sequential relationships in the data with fewer parameters and faster training times compared to LSTM.

A. Motivation

The hindrances to post quality and user experience faced by these popular question answers motivated us to develop an automated system to assist in moderation. This will effectively reduce the load on human moderators and reduce redundancy, leading to faster solutions to problems which in turn would ultimately empower newcomers in technical fields, enabling them to engage in discussions more effectively.

B. Objective

The objective of this research is to explore the relationship between the content of the post and its quality by utilising machine learning and deep learning approaches to train a model which can classify the quality of a post before the user posts it. Deep learning models like BiLSTM, GRU, along with BERT and its variants, DistilBERT, ALBERTa, and RoBERTa were used for training and evaluation of the Stackoverflow post data provided by ’60k Stack Overflow Questions with Quality Rating’ posted on Kaggle Inc., which was contributed by Moore.

C. Paper Structure

This research paper has been branched into six sections. Section I highlights the introduction which gives a brief idea about this research along with the definition of the objective and the motivation behind this study. Section II provides the review of related literature and methods that were used in this study. Section III is the methodology section that discusses about the various steps involved in obtaining the desired results which are then discussed in the fourth section. Finally, the fifth and final section concludes the research discussing about the key findings and future scope of this research.

II. Literature Review

This paper presents a combined model that joins Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to boost the classification of Stack Overflow post quality [10] . Regular machine learning models often struggle to capture both structural and sequential links in text data. To fix this, the suggested method uses CNN to capture relevant features and LSTM to spot sequential dependencies. The model looks at 20 features split into text properties, readability, formatting, and community interaction. It reaches an accuracy of 80.45% beating classic classifiers like SVM, KNN, MLP. The model’s performance is checked using basic classification metrics, on a public Stack Overflow dataset. Unlike earlier work that used binary classification, this model sorts posts into four quality levels making a more detailed breakdown. The model has some weak points such as uneven classes and lower precision/recall scores. These issues suggest room to improve by tweaking parameters adjusting features, and looking into more complex deep neural network structures. This research investigates the application of deep learning methods to detect duplicate questions in Stack Overflow with the help of Word2Vec embeddings to promote semantic comprehension [11]. It proposes three models, by merging Word2Vec embeddings with CNN, RNN, and LSTM frame- works. The dataset is drawn from a collection with duplicate and non-duplicate question pair data over six programming languages, from the Stack Overflow Data Dump (2008–2014). The models are tested through various metrics. Experimental results show that WV-CNN and WV-LSTM perform better than current baseline models, classical machine learning mod- els, and existing deep learning methods. The work indicates the usefulness of combining Word2Vec with deep learning to detect and rank duplicate questions.

This work investigates the application of neural network models in measuring the quality of questions asked on Stack Overflow, a popular programming Q&A website [12]. It is a comparison between the efficiency of deep learning methodologies and the classical models, which illustrates that the application of neural networks offers more precision in classifying high-quality and low-quality questions.

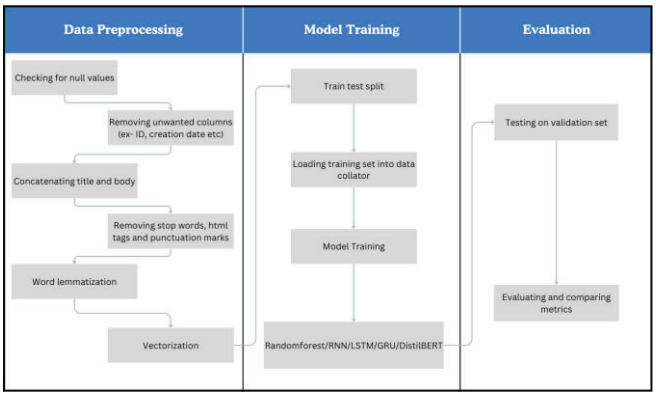

Fig. 1. Process Flow Chart

This study proposes a new way of evaluating question quality in Stack Overflow using the question language rather than the user metadata or interaction[13]. They use Ridge regression models on a 1.7 million post dataset to investigate how some lexical features—expressions of emotion, technical issues, or misspellings—affect the score and number of answers provided for a question. The findings indicate and mark that the linguistic characteristics alone can be accurately predict question quality, with their largest model being the best. By incorporatingly joining semantic and part-of-speech analysis, the work highlights the importance of question wording, offering real-world evidence of writing style contributing

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to increased engagement. Hence, the study offers a language-centered approach to predicting quality and calls out future research into linguistic signals in CQA environments.

This study addresses the issue of duplicate questions on Stack Overflow, which can delay answers and burden moderators[14]. Through manual analysis, the authors identify key reasons for duplication, such as unclear phrasing and varied terminology. They propose a classification-based method that uses selected features to detect duplicates, showing improved accuracy and higher recall compared to the existing DupPredictor model. The findings highlight the value of language-based features in automating duplicate detection and enhancing platform efficiency.

This study tackles the problem of predicting the quality of the questions on Stack Overflow as a text classification task, aiming to reduce low quality posts and ease moderate efforts[15]. The authors preprocess and vectorize question posts, then train five machine learning models and two deep learning models (Bi-LSTM and BERT) for comparison. Experimental results show that model performance varies significantly with parameter tuning, with accuracies ranging from 54% to 92%. The findings highlight the effectiveness of attention-based models like BERT for quality prediction in Q&A platforms.

III. METHODOLOGY

This section highlights the steps implemented in our approach, encompassing the various stages of text preprocessing, tokenization, vectorization, and the deep learning models utilised. The approach follows a structured pipeline that begins with the preparation of raw text data and progresses through multiple stages, including data cleaning, transformation and feature extraction. In Figure 1 the process flow chart reflects the whole procedure of this study.

A. Dataset

The dataset utilized in this research is sourced from the ’60k Stack Overflow Questions with Quality Rating’ collection available on Kaggle, contributed by Moore(10). It consists of a comprehensive set of Stack Overflow posts, each associated with various features, including the Post ID, title, body, relevant tags and a creation date. The posts are categorized based upon their quality into three classes: HQ, LQ close and LQ edit. In total, 60000 questions were used for model training.

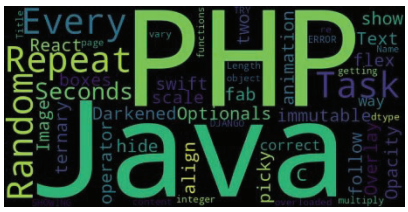

Fig. 2. Word Cloud

The words that appear most frequently in the ”Title” section of the dataset are shown graphically in Figure 2.

Figure 3 shows the length of posts in words by classifying them in 3 main categories, i.e. short, medium and long.

Figure 4 indicates the distribution of data for each of the 3 labels and the equal division clearly suggests that the data is free of class imbalance.

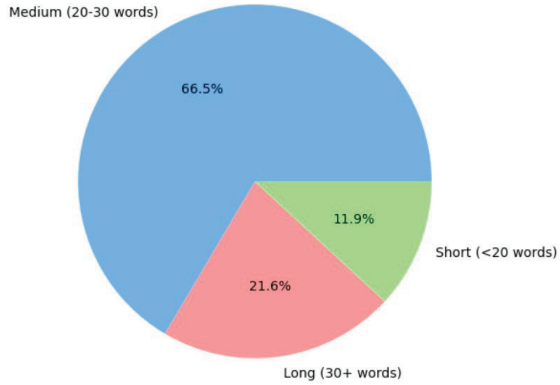

Fig. 3. Post Length

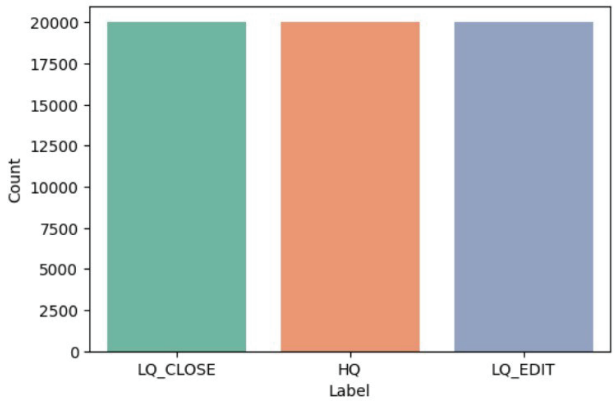

Fig. 4. Value Counts

B. Preprocessing

In the preprocessing phase, a series of systemic steps were employed to prepare the dataset for model input. These steps ensured the text data was clean, standardized and formatted uniformely to be used for the classification task.

1) Null Value Removal:

To maintain data integrity and ensure consistent input, records with missing or null values were first detected and then eliminated from the dataset.

2) Removal of Unnecessary Columns:

Irrelevant columns such as the post ID, post creation date and tags associated with the posts were excluded from the dataset. These attributes did not contribute to the classification task and would introduce unnecessary noise into the results. Only the post title and body text, containing the most relevant information, were utilised.

3) Concatentation of Relevant columns:

The title and body content of the post were concatenated into a single string, ensuring the model had access to the full context of the post in a unified form. This step allowed the model to better understand the relation between the post title and content of each post, leading to accurate predictions.

4) Stopword Removal:

To further refine the dataset, com- mon stopwords-words contributing little semantic weight (”the”,”and”,”for”)-were removed. This reduction simplified the input and allowed the model to focus and concentrate on more meaningful and informative terms, which were essential for better classification.

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5) HTML entity removal:

Given that the data was originally scraped from StackOverflow, the text data contained multiple HTML entities (”<p>”,”&”), these were removed from the text to preserve only the textual data from the scraped data. Removal of HTML tags ensured that the models processed clean text without any irrelevant markup.

6) Punctuation Removal:

Punctuation marks were also removed during the preprocessing as punctuations do no typically contribute much meaning to the text in this context. Eliminating it helped in minimizing noise and improve model performance.

7) Lemmatization:

To ensure consistency and enhance the model’s understanding of the semantic meaning of the words, a lemmatization process was utilised. Lemmatization reduces the word to its base form (”running” to ”run”), which helps in standardizing the text and reduce variations of the same word.

8) Tweet Preprocessing:

Special tweet preprocessing methods were also applied to handle unique nature of text similar to tweets. This included removal of unique tweet entities like URLs, hashtags, etc. which are often present in informal textual content but are irrelevant in the context of this classification task.

These preprocessing steps were necessary in transforming the raw text into a clean and structured represenatation suitable for training the transformer based models. By systematically addressing potential causes of noise and inconsistency, the preprocessing pipeline ensured that the models focus only on the key features of the data and not on the noise.

C. Model Training

In the model training phase, we employed a suite of six deep learning models to perform the quality classifica- tion of StackOverflow posts. These comprised of two re- current neural network (RNN) architectures-Long Short-term Memory (LSTM) and Gated Recurrent Unit (GRU)-alongside four transformer based models-BERT, DistilBERT, ALBERT, RoBERTa-sourced from Hugging Face Hub. The selections of the models were guided by the findings presented in the Literature Review, where their respective strengths in handling textual data are discussed. The preprocessed text was tokenized using Hugging Face Hub’s AutoTokenizer, which converted the input into token IDs, attention masks, and applied appropri- ate padding to maintain uniform input length. Label encoding was also applied to convert categorical target labels into a compatible numerical input format. All models were fine- tuned by enabling gradient updates across all layers, allowing for better adaptation to the StackOverflow data. Evaluation of model performance was conducted using classification metrics, which include F1-score, precision, accuracy, recall, to provide a comprehensive result across different aspects of predictions quality. Upon completion of training, each model was evaluated on the validation set, and the best performing models were subsequently uploaded to Hugging Face Model Hub to ensure reproducibility and facilitate deployment.

IV. Results & Discussion

As stated above, we have performed training/fine-tuning of two neural network models and four transformer-based models, then we applied hyperparameter tuning and observed the changes in the metrics of the models. The results of our study are represented in Table 1, depicting a direct comparison between all of the above-mentioned models in terms of their accuracy, precision, recall, and f1-scores. Figure 5 shows the comparison visually for better understanding.

TABLE I. Evaluation Results

Model	Accuracy	Precision	Recall	F1-score
DistilBERT	90.66%	0.92	0.92	0.91
RoBERTa	86.54%	0.88	0.88	0.88
ALBERT	88.68%	0.90	0.90	0.90
BERT	93.52%	0.94	0.94	0.94
LSTM	86,88%	0.86	0.86	0.86
GRU	87.4%	0.87	0.87	0.87

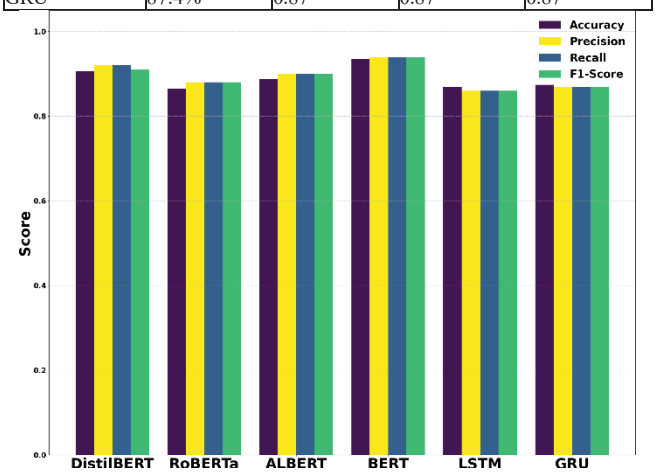

Fig. 5. Performance Comparison Graph

Table I and Figure 5 clearly shows that the BERT approach surpasses others in terms of the accuracy, precision, recall and F1-score.This superior performance is explained by BERT’s deeper architecture and bidirectional attention mechanism, which allows it to capture semantic relationships better than lighter or distilled versions like distilBERT, RoBERTa or ALBERT. The ability to capture these relationships is crucial when it comes to problems like analyzing and predicting questions quality on StackOverflow as context plays a vital role in determining the quality.

TABLE II. Comparison With Other Related Works

Paper	Approach	Accuracy	Recall	Precision	F1-Score
[16]	Deep learning with NLP	74.0%	74.0%	75.0%	-
[17]	Bi-LSTM and CNN models	85.0%	97.0%	-	93.0%
Our Result	BERT-base	93.0%	94.0%	94.0%	94.0%

Table II shows the comparison of our results with two different studies that were conducted on the same problem statement. The authors of [16] utilized a deep learning com- bined with natural language processing approach, out of the four experiments that they performed by carefully changing the sample size and hyperparameter tuning, they provided their final results as discussed in the table. Whereas the author of [17] used two approaches, a bi-LSTM model and a CNN model which were trained on the same dataset used in our study.The concluding results are showcased in Table II. Our best result indicates that our model surpasses the above stated papers in terms of these evaluation metrics which clearly

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shows that transformer-based models like BERT surpass the traditional approaches.

V. Conclusion And Future Work

As the popularity of platforms such as Stack Overflow continues to grow, user-generated content quality assurance has become a major concern. With billions of questions being posted, not much manual moderation seems plausible any-more. The proposed research focused on using deep learning, especially transformer-based models like BERT, to automate the question quality classification process.

Among all the models tested, BERT-base gave the best result in all scenarios, which demonstrates the benefit of using trans- former architecture understanding contextual language. The results in Table 1 do justify that we have made the right choice. This work appears to present an avenue for the application of transformer-based models in maintaining the quality of large- scale Q&A platforms. Future works could look into using extra metadata, employing larger model variants in experimentation, or even extending to the support of multilingual data for increased robustness and range of applications.

While this research demonstrated the effectiveness of transformer-based models like BERT for predicting quality of the questions, there are more ways to make this approach even stronger.

1) By incorporating real time data for the model we can boost its performance and provide more context. The model can be included as a live moderation tool to help in the moderation process of flagging potentially low- quality questions in real-time which will enhance user experience high quality content.

2) In addition, obtaining better and more diverse datasets that include a vast amount of question topics and styles may enable the model to generalize more, enhancing its robustness and minimizing potential overfitting.

3) Compressing the model would also prove really effective to make it computationally less expensive which would help in real-world deployment on low resource environments and faster model turnaround time.

While the study shows satisfactory results, there are still some limitations. The size of the dataset can be one of the key limitations of this research which limits the amount of context the model can uncover so there can be some discrepancies in the predicted quality. This can be fixed by utilizing data dumps provided by stackoverflow.

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